

Localized buckling in sandwich struts with inhomogeneous deformations in both face plates

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Abstract

A nonlinear analytical model for investigating localized interactive buckling in simply supported thin-face plate sandwich struts with weak cores is extended to account for local deformations in both face plates, which have been observed in experiments and finite element simulations. The original model is based on potential energy principles with large displacement assumptions. It assumes Timoshenko shear deformable theory for the core and approximates the overall mode as a half-sine wave along the length of the strut while the local face plate displacements are initially unknown and are found as solutions of the governing equations. The extended model is able to capture measurable local face plate displacements in the less compressed face plate, beyond the secondary bifurcation which leads to localized interactive buckling, for the case where overall buckling is critical. Moreover, the allowance of local displacements in both face plates allows the extended model to predict the post-buckling behaviour better in cases where local buckling is critical. The results from this model compare very well with nonlinear finite element simulations with respect to both the equilibrium paths and panel deformations.

Keywords: Mode interaction, Sandwich struts, Variational formulation, Interactive buckling, Post-buckling

1. Introduction

Sandwich construction is well-known in many engineering disciplines as a versatile structural solution with a favourable combination of low weight and high bending stiffness [1]. Such structures usually comprise two stiff thin plates separated by a soft compliant core. By positioning the stiff plates away from one another increases the bending rigidity of the sandwich structure, while the compliant core is able to resist shear stresses. This optimized configuration, however, is prone to instability phenomena such as overall buckling

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and wrinkling. These phenomena are triggered at certain critical loads with sandwich structures buckling either globally (effectively as an Euler strut) or locally in each face plate in a symmetric (hourglass) or antisymmetric (snake-like) fashion, as shown in Figure 1 [2, 3]. Moreover these types of instability might interact leading to localized buckling [4, 5], or bring about damage in terms of face–core delamination [6, 7], both of which can destabilize the structural component violently. Since imperfections can have an erosive effect on the critical load carrying capacity of the sandwich strut [8], it is very important for these effects to be accounted for when modelling these components properly.

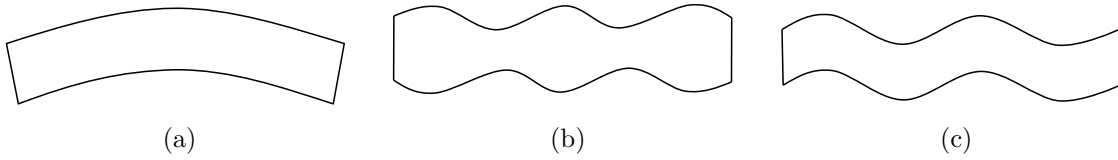


Figure 1: The three principal buckling modes of sandwich strut. (a) Overall or global buckling; (b) symmetric or the hourglass mode of local face plate buckling; (c) antisymmetric or the snake mode of local face plate buckling.

Critical loads for overall buckling and wrinkling (local buckling) in sandwich panels have been reported in the classical texts of Allen [9] and Plantema [10]. Fazio [11] and Drysdale [12] reported on different models for various loadings on sandwich structures. Frostig and co-workers [13, 14] provided elasticity solutions for different structural configurations while analytical and numerical models for overall buckling and wrinkling have been studied extensively [15, 16, 17, 18, 19], including multiscale techniques by Yu et al [20]. Léotoing et al [21] investigated the geometrically nonlinear interaction between overall and local buckling modes, which were first successfully captured analytically by Hunt and Wadee [4] using a variational formulation based on the potential energy method and Timoshenko bending theory. Further work by Wadee and co-workers included the extension of the interactive buckling model for orthotropic cores [22], higher order shear deformable theories [5], beam-columns [23] and functionally graded cores [24]. The development of using novel materials in sandwich construction such as metal foams for the core and high strength composites for the face plates has led to renewed interest in investigating the stability of such structures with metal foam cores [25, 26] and for composite reinforced sandwich structures [27].

Mode interaction and localization in sandwich structures is a very important phenomenon since it can dominate the post-critical response, regardless of the initiating mode [8]. This work seeks to extend the fundamental model for interactive buckling developed by Hunt and Wadee [4] which considers localization that occurs solely at the more compressed face of the sandwich strut. More recently, Wadee and Bai [28] successfully modelled interactive buckling between minor axis buckling of an I-section strut, using a similar modelling technique. Although this is a different problem, the aim herein is to present an analytical model that investigates and accounts for local deformations on both face plates during post-buckling. Specifically for the case where overall buckling occurs first, it has been

observed both experimentally [29] and in FE simulations [5] (see Figure 2) that smaller amplitude local deformations begin to appear practically immediately after the secondary bifurcation when local buckling within the other face plate has been triggered. The current work aims to capture these responses and study their effects on the post-buckling response of different sandwich configurations. The results from this model are compared with the results obtained from Hunt and Wadee [4] as well as FE simulations performed in the commercial finite element software ANSYS [30]. Further considerations and extensions of the model are subsequently discussed.

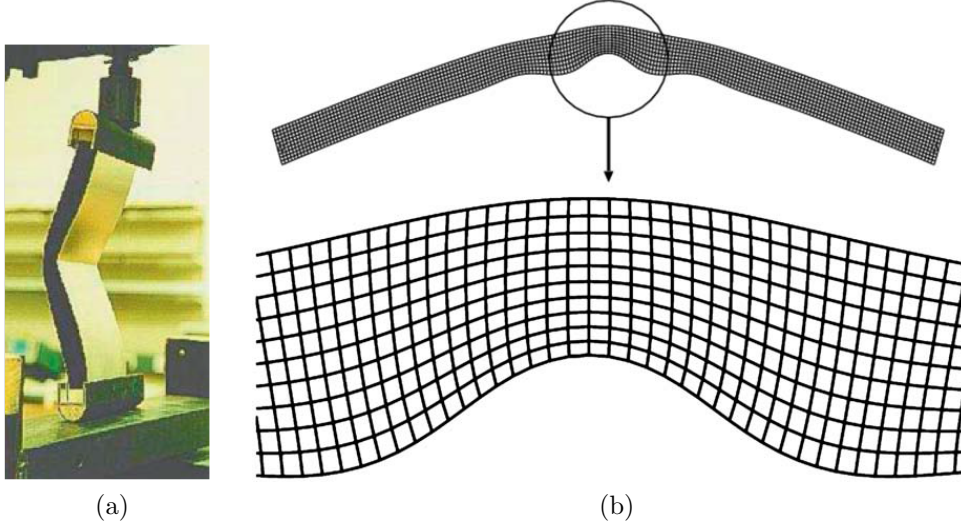


Figure 2: Interactive localized buckling exhibited in the more compressed face plate and local deformations within the less compressed face plate as exhibited in (a) experimental work [29] and (b) FE simulations performed in ABAQUS [5].

2. Analytical formulation

2.1. Definitions, Geometry and Kinematics

In the current work, the original Hunt and Wadee model [4]—hereinafter termed the ‘1998 model’—is modified to allow local deformations within both face plates. The dimensions of the sandwich strut and the coordinate system are shown in Figure 3. The model assumes isotropic face plates with Young’s modulus E and Poisson’s ratio ν as well as an isotropic core material with Young’s modulus E_c and Poisson’s ratio ν_c . Loading is applied axially by a compressive force P acting at the mid-section, through rigid end plates to ensure equal load transfer to the face plates.

The purely compressive displacement prior to buckling is represented by ΔL , while overall (Euler-type) buckling is decomposed into sway and tilt components, $W(x)$ and $\theta(x)$ (see Figure 4). These components are represented by predetermined trigonometric functions with dimensionless amplitudes q_s and q_t respectively as:

$$W(x) = q_s L \sin \frac{\pi x}{L}, \quad \theta(x) = q_t \pi \cos \frac{\pi x}{L}, \quad (1)$$

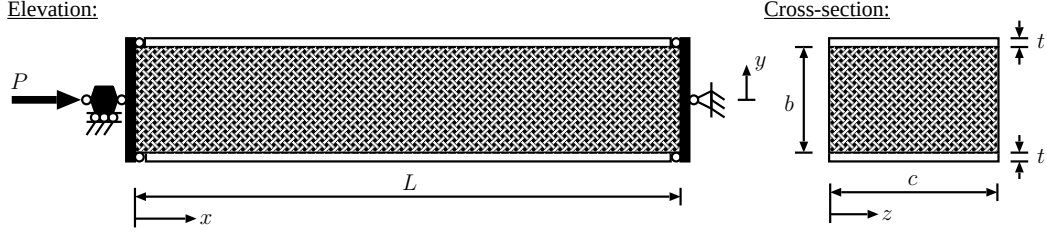


Figure 3: A sandwich strut of length L , core thickness b , face plate thickness t , under an axial load P . The breadth of the strut out of the plane shown is c . The coordinate system is also presented in the longitudinal (x) and transverse (y) directions. The z -axis across the breadth of the strut is orthogonal to the xy plane.

since it is well known that the solution for the overall buckling mode of a simply supported strut can be best approximated by a half-sine wave and the total shear effect is determined by the difference of sway and tilt. The amplitudes q_s and q_t , as well as Δ enter the model as generalized coordinates, which are computed during the solution process. The new model retains the Timoshenko beam theory assumption for the core material under which shear strains are allowed to develop within the core, which are essential for including the interaction between the two buckling modes of different length scales [5]. For the range of application of this model (weak, isotropic core and stiff thin face plates) Timoshenko shear deformable theory for the core suffices and offers a sufficiently good approximation of the kinematics of the overall mode. Furthermore, along with the contributions of the sway and tilt components for overall buckling to the displacement of the strut, both face plates are allowed to form local deformations. Hence, for the top face plate $w_t(x)$ is defined as the displacement of the buckled face plate perpendicular to the unbuckled face, and $u_t(x)$ is defined as the displacement parallel to the unbuckled face. The corresponding displacements for the bottom face plate are defined as $w_b(x)$ and $u_b(x)$. These functions of x have no predetermined form and are sought as solutions from the minimization of the total potential energy functional formulated in a later subsection. To ensure continuity of the displacements within the core, the top and bottom interactive mode displacements are taken to vary linearly with y , as:

$$w_c(x, y) = \left(\frac{b + 2y}{2b} \right) w_t(x) + \left(\frac{b - 2y}{2b} \right) w_b(x), \quad (2)$$

$$u_c(x, y) = \left(\frac{b + 2y}{2b} \right) u_t(x) + \left(\frac{b - 2y}{2b} \right) u_b(x), \quad (3)$$

so that pure deformations for each face plate can be easily extracted as the solutions of the governing differential equations, while superposition can be used to quantify the displacement due to local deformations within the core. The linear variation of each displacement function across the depth of the core was deemed appropriate, since the magnitudes of these displacements are generally small and previous trials with nonlinear variations did

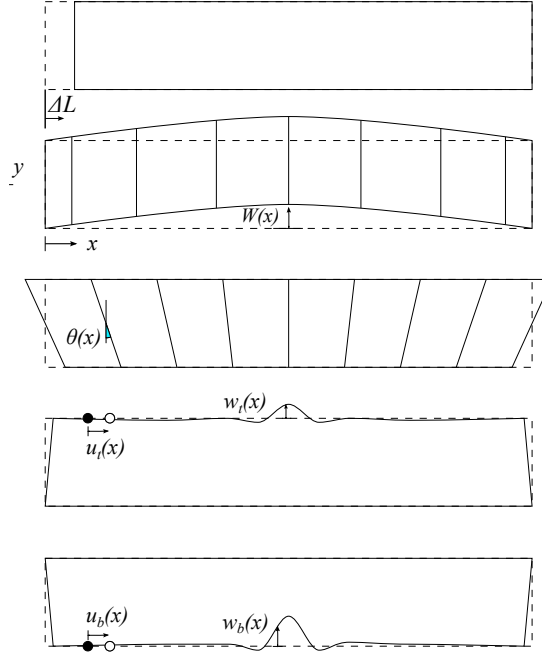


Figure 4: From top to bottom: Purely compressive displacement prior to buckling denoted by ΔL ; overall buckling sway mode $W(x)$; overall tilt mode $\theta(x)$; top face plate transverse $w_t(x)$ and in-plane $u_t(x)$ displacements and bottom face plate transverse $w_b(x)$ and in-plane $u_b(x)$ displacements.

not reveal any significant changes in the response.

2.2. Strain Energy

As in the original work by Hunt and Wadee [4], a total potential energy approach with large displacement assumptions is followed to arrive at the equilibrium equations for the strut. The total potential energy V of the system comprises the strain energy of the sandwich strut integrated over its entire volume minus the work done by the applied load. In sandwich struts, the face plates are considered to provide the principal resistance to bending. The primary function of the core, on the other hand, is to resist shear deformation and does not intrinsically provide significant bending or axial resistance. Hence, the total strain energy of the strut is compartmentalized into strain energy components associated with pure bending due to buckling, membrane stretching of the face plates with core shear and transverse deformation. Finally, the work done by the applied load is introduced in the potential energy functional. Assuming a plane stress condition, the potential energy of the strut is readily integrated over the breadth of the strut. The procedure for formulating the total energy is thoroughly explained in Hunt & Wadee [4] with additional relevant discussions provided in Yiatros et al [24].

2.3. Bending Energy

Presently, the strain energy due to the overall and local curvature on the face plates is considered. These are taken as their linearized form; the second derivative of global

and local lateral displacements, since the bending strains are small and the cross terms add complexity to the formulation without changing the response significantly. The strain energy from the local and global curvature is assembled, neglecting cross terms that play a negligible role in the total potential energy functional for a sandwich panel geometry where $b \gg t$ and $L \gg b$ or $t \ll b$ and $b \ll L$:

$$U_b = \frac{1}{2}EI \int_0^L (2W''^2 + w_t''^2 + w_b''^2) \, dx, \quad (4)$$

where $EI = Ect^3/[12(1 - \nu^2)]$ and primes indicate differentiation with respect to x .

2.4. Membrane energy

Along with bending energy, face plates are also subjected to membrane action. Unlike the previous model, axial strains include global and local contributions for both face plates, as shown below:

$$\begin{aligned} \varepsilon_{x,t} &= -\frac{b}{2}\theta' - \Delta + \frac{1}{2}w_t'^2 + u_t', \\ \varepsilon_{x,b} &= \frac{b}{2}\theta' - \Delta + \frac{1}{2}w_b'^2 + u_b', \end{aligned} \quad (5)$$

where u_b , u_t , w_b and w_t are all functions of x only given on the right hand sides of the expressions in Equations (2) and (3). The nonlinear terms for each face plate arise from Von Kármán plate theory. Hence the membrane energy is assembled as follows:

$$\begin{aligned} U_m &= D \int_0^L (\varepsilon_{x,t}^2 + \varepsilon_{x,b}^2) \, dx, \\ &= D \int_0^L \left[2\Delta^2 + \frac{1}{2}b^2\theta'^2 + u_t'^2 + u_b'^2 + \frac{1}{4}(w_t'^4 + w_b'^4) \right. \\ &\quad \left. + (b\theta' - 2\Delta) \left(u_b' - u_t' + \frac{1}{2}w_b'^2 - \frac{1}{2}w_t'^2 \right) + u_t'w_t'^2 + u_b'w_b'^2 \right] \, dx, \end{aligned} \quad (6)$$

where $D = Ect/2$.

2.5. Core strain energy

As in the 1998 model [4], the strain energy stored in the core, U_c is derived from considering shear and transverse displacements only, whereas the longitudinal strains are omitted for simplicity since they do not play a major role in the resistance when compared to the role of the face plates. This derives from the assumption that the core is soft and hence the strain energy stored in the system is very small compared to the face plate membrane energy. A subsequent model presented in [22], which accounted for the longitudinal strain energy in the core, had an overall buckling critical load less than 1% different from the current formulation, while the position of the secondary bifurcation and interactive buckling deformations were virtually identical. It was thus deemed appropriate to neglect the

contributions from direct, longitudinal strains from the core strain energy, the complexity of the model is reduced, without losing significant accuracy.

$$U_c = \frac{1}{2} \int_0^c \int_{-b/2}^{b/2} \int_0^L \left(\frac{E_c}{1 - \nu^2} \varepsilon_y^2 + G_c \gamma_{xy}^2 \right) dx dy dz, \quad (7)$$

where $G_c = E_c/[2(1 + \nu_c)]$ and

$$\varepsilon_y = \frac{w_t(x) - w_b(x)}{b}, \quad (8)$$

$$\gamma_{xy} = W'(x) - \theta(x) + \left(\frac{b + 2y}{2b} \right) w'_t(x) + \left(\frac{b - 2y}{2b} \right) w'_b(x) + \frac{u_t - u_b}{b}. \quad (9)$$

The core strain energy can be readily integrated over the breadth and depth of the core, resulting in an integral over the length. Thus:

$$\begin{aligned} U_c = \int_0^L \left\{ G \left[\frac{w_t'^2 + w_b'^2 + w'_t w'_b}{3} + \frac{u_t w'_b + u_t w'_t - u_b w'_b - u_b w'_t}{b} \right. \right. \\ \left. \left. + \frac{u_t^2 + u_b^2 - 2u_t u_b}{b^2} + (W' - \theta) \left(w'_t + w'_b - \frac{2}{b} (u_b - u_t) \right) \right. \right. \\ \left. \left. + (W' - \theta)^2 \right] + \frac{1}{2} k (w_t - w_b)^2 \right\} dx. \end{aligned} \quad (10)$$

where $G = G_c bc/2$ and $k = E_c c/[b(1 - \nu_c^2)]$.

2.6. Work done by applied load

The work done by the external force is the product of the applied load P , multiplied by the total displacement $\mathcal{E}$ at the point of application, i.e. at the mid-plane of the strut. The procedure for obtaining the total displacement at the point of application has been described in Yiatros et al [24]. For the kinematic variables in the current model, the work done by the external force has the form:

$$P\mathcal{E} = P \int_0^L \left[\frac{1}{2} W'^2 + \Delta - \frac{1}{2} (u'_t + u'_b) \right] dx, \quad (11)$$

where the first term is the contribution from the overall buckling lateral displacement W if the strut is inextensional. The second term arises due to the uniform compressive strain Δ which gives the non-trivial pre-buckling end shortening. The remaining terms are the contributions of the localized displacement components u_t and u_b of the top and bottom face plates within the core. The contribution due to these components is from the relative movement at the two end-points.

2.7. Total potential energy, critical load and equilibrium equations

The total potential energy V of the strut is obtained by summing all the energy contributions, such that:

$$\begin{aligned}
V &= U_b + U_m + U_c - P\mathcal{E}, \\
&= \int_0^L \left\{ \frac{EI}{2} \left(2q_s^2 \frac{\pi^4}{L^2} \sin^2 \frac{\pi x}{L} + w_t''^2 + w_b''^2 \right) + D \left[\frac{1}{2} b^2 \frac{q_t^2 \pi^4}{L^2} \sin^2 \frac{\pi x}{L} \right. \right. \\
&\quad + 2\Delta^2 + u_t'^2 + u_b'^2 + \frac{1}{4} (w_t'^4 + w_b'^4) + u_t' w_t'^2 + u_b' w_b'^2 \\
&\quad + \left. \left(\frac{q_t b \pi^2}{L} \sin \frac{\pi x}{L} - 2\Delta \right) \left(u_b' - u_t' + \frac{1}{2} w_b'^2 - \frac{1}{2} w_t'^2 \right) \right] \\
&\quad + G \left[(q_s - q_t)^2 \pi^2 \cos^2 \frac{\pi x}{L} + \frac{1}{3} (w_t'^2 + w_b'^2 + w_b' w_t') \right. \\
&\quad + (q_s - q_t) \left(w_t' w_b' + \frac{2u_t - 2u_b}{b} \right) \pi \cos \frac{\pi x}{L} + \frac{u_t^2 + u_b^2 - 2u_b u_t}{b^2} \\
&\quad + \left. \frac{u_t w_b' + u_t w_t' - u_b w_b' - u_b w_t'}{b} \right] + \frac{1}{2} k (w_t - w_b)^2 \\
&\quad \left. - P \left(\frac{1}{2} q_s^2 \pi^2 \cos^2 \frac{\pi x}{L} - \frac{1}{2} u_t' - \frac{1}{2} u_b' + \Delta \right) \right\} dx.
\end{aligned} \tag{12}$$

Note that by setting all w_t and u_t terms and their derivatives to zero, the potential energy functional becomes identical to the 1998 model. Hence the critical load for overall buckling, given in Equation (13) remains the same as in [4]:

$$P^C = \frac{2\pi^2 EI}{L^2} + \frac{Gb^2 \pi^2}{L^2} \left[\frac{2D}{2G + Db^2 \pi^2 / L^2} \right]. \tag{13}$$

The governing nonlinear equations of equilibrium are found by minimizing the total potential energy functional. For the system to be in equilibrium, the potential energy has to be stationary with respect to the functions w_b , w_t , u_b and u_t along with the generalized coordinates q_s , q_t and Δ . The stationary point of the functional is computed through a hybrid formulation; the calculus of variations [31] is used for computing the stationary points with respect to the unknown functions w_b , w_t , u_b , u_t , whereas the stationary points with respect to the generalized coordinates are computed as in the classical Rayleigh–Ritz method by setting the partial derivatives of the energy functional with respect to q_s , q_t and Δ to zero. This leads to four nonlinear ordinary differential equations (ODEs) that are 4th order with respect to w_t and w_b and 2nd order with respect to u_t and u_b , plus three

integral constraints, given in Equations (14)–(20):

$$\begin{aligned}
EIw_b'''' + D \left[2\Delta w_b'' + q_t \frac{b\pi^2}{L} \left(\sin \frac{\pi x}{L} w_b'' + \frac{\pi}{L} \cos \frac{\pi x}{L} w_b' \right) - 3w_b'^2 w_b'' \right. \\
\left. - 2w_b'' u_b' - 2w_b' u_b'' \right] + G \left[\frac{u_b' - u_t'}{b} - \frac{2w_b'' + w_t''}{3} + (q_s - q_t) \frac{\pi^2}{L} \sin \frac{\pi x}{L} \right] \\
+ k(w_b - w_t) = 0,
\end{aligned} \tag{14}$$

$$\begin{aligned}
EIw_t'''' + D \left[2\Delta w_t'' + q_t \frac{b\pi^2}{L} \left(\sin \frac{\pi x}{L} w_t'' + \frac{\pi}{L} \cos \frac{\pi x}{L} w_t' \right) - 3w_t'^2 w_t'' \right. \\
\left. - 2w_t'' u_t' - 2w_t' u_t'' \right] + G \left[\frac{u_b' - u_t'}{b} - \frac{2w_t'' + w_b''}{3} + (q_s - q_t) \frac{\pi^2}{L} \sin \frac{\pi x}{L} \right] \\
+ k(w_t - w_b) = 0,
\end{aligned} \tag{15}$$

$$\begin{aligned}
u_b'' + w_b' w_b'' - \frac{b\pi^3 q_t}{2L^2} \cos \frac{\pi x}{L} + \frac{G}{Db} \left(\frac{u_t - u_b}{b} + \frac{w_t' + w_b'}{2} \right. \\
\left. + (q_s - q_t) \pi \cos \frac{\pi x}{L} \right) = 0,
\end{aligned} \tag{16}$$

$$\begin{aligned}
u_t'' + w_t' w_t'' + \frac{b\pi^3 q_t}{2L^2} \cos \frac{\pi x}{L} + \frac{G}{Db} \left(\frac{u_b - u_t}{b} - \frac{w_b' + w_t'}{2} \right. \\
\left. - (q_s - q_t) \pi \cos \frac{\pi x}{L} \right) = 0,
\end{aligned} \tag{17}$$

$$\begin{aligned}
P = \frac{2G}{q_s} \left\{ q_s - q_t + \frac{1}{\pi L} \int_0^L \left[\left(w_b' + w_t' - \frac{2u_b - 2u_t}{b} \right) \cos \frac{\pi x}{L} \right] dx \right\} \\
+ \frac{2EI\pi^2}{L^2},
\end{aligned} \tag{18}$$

$$\begin{aligned}
q_s = q_t \left(1 + \frac{Db^2\pi^2}{2GL^2} \right) + \int_0^L \left[\frac{1}{\pi L} \left(\frac{2u_b - 2u_t}{b} - w_b' - w_t' \right) \cos \frac{\pi x}{L} \right. \\
\left. - \frac{Db}{GL^2} \left(u_b' - u_t' + \frac{w_b'^2}{2} - \frac{w_t'^2}{2} \right) \sin \frac{\pi x}{L} \right] dx,
\end{aligned} \tag{19}$$

$$\Delta = \frac{P}{4D} + \frac{1}{2L} \int_0^L \left(u_t' + u_b' + \frac{1}{2} w_b'^2 + \frac{1}{2} w_t'^2 \right) dx. \tag{20}$$

The governing equations are subject to the following boundary conditions that minimize V :

$$w_b(0) = w_b(L) = w_b''(0) = w_b''(L) = 0, \tag{21}$$

$$w_t(0) = w_t(L) = w_t''(0) = w_t''(L) = 0, \quad (22)$$

$$u_b'(0) + \frac{1}{2}w_b'^2(0) - \Delta = u_t'(0) + \frac{1}{2}w_t'^2(0) - \Delta = -\frac{P}{4D}, \quad (23)$$

$$u_b'(L) + \frac{1}{2}w_b'^2(L) - \Delta = u_t'(L) + \frac{1}{2}w_t'^2(L) - \Delta = -\frac{P}{4D}, \quad (24)$$

which ensure simple supports for face plates and matching in-plane stresses at both ends.

3. Nonlinear stability analysis

In order to establish the non-trivial interactive modes, the system of governing ordinary differential equations, integral constraints and boundary conditions was nondimensionalized and solved using AUTO [32], the powerful numerical continuation software with the ability to trace evolving solutions with parametric changes which allow to plot the post-buckling equilibrium paths. The chosen leading parameter—usually P in the current case—is varied and AUTO solves for the remaining parameters (q_s , q_t and Δ) as well as the unknown functions and their derivatives. The investigation of buckling mode interaction is conducted for the cases where either overall or local buckling is critical. The chosen leading parameter is varied from zero along the fundamental path until a bifurcation point is found. The load parameter in AUTO is normalized with respect to the overall buckling load, so that when overall buckling is critical, $p = P/P_O^C = 1$. A second run is then performed from the primary bifurcation until the secondary bifurcation, beyond which interactive buckling is triggered in the bottom—more compressed—face plate, while simultaneously, deformations begin to appear at the top face plate. In the next few subsections some key results from the analysis described above are presented and discussed.

3.1. Results

A typical sandwich strut configuration comprising stiff aluminium alloy plates and a polymeric compliant foam was used. The material and geometrical properties are given below:

Face plate Young's modulus, $E = 68,947.57 \text{ N/mm}^2$,
Poisson's ratio, $\nu = 0.3$,
Core Young's modulus, $E_c = 198.57 \text{ N/mm}^2$,
Poisson's ratio, $\nu = 0.2$,
Core Shear modulus, $G_c = 82.74 \text{ N/mm}^2$,
Face plate thickness, $t = 0.508 \text{ mm}$,
Core depth, $b = 5.08 \text{ to } 18 \text{ mm}$,
Strut length, $L = 100 \text{ to } 200 \text{ mm}$.

The same material properties and some of the geometric ones have been used in earlier relevant publications [3, 4] and are also used currently for comparison purposes. The materials are considered to be linearly elastic for the ranges of strains studied. The geometrical properties were selected to include cases of criticality of the overall mode and others of the local (face plate) modes.

3.1.1. Critical overall buckling

Considering the first strut, where $b = 5.08$ mm and $L = 100$ mm, overall buckling is critical once the secondary bifurcation is located the strut subsequently destabilizes, rapidly losing its load carrying capacity as seen in Figure 5(a). It can be seen that by comparing

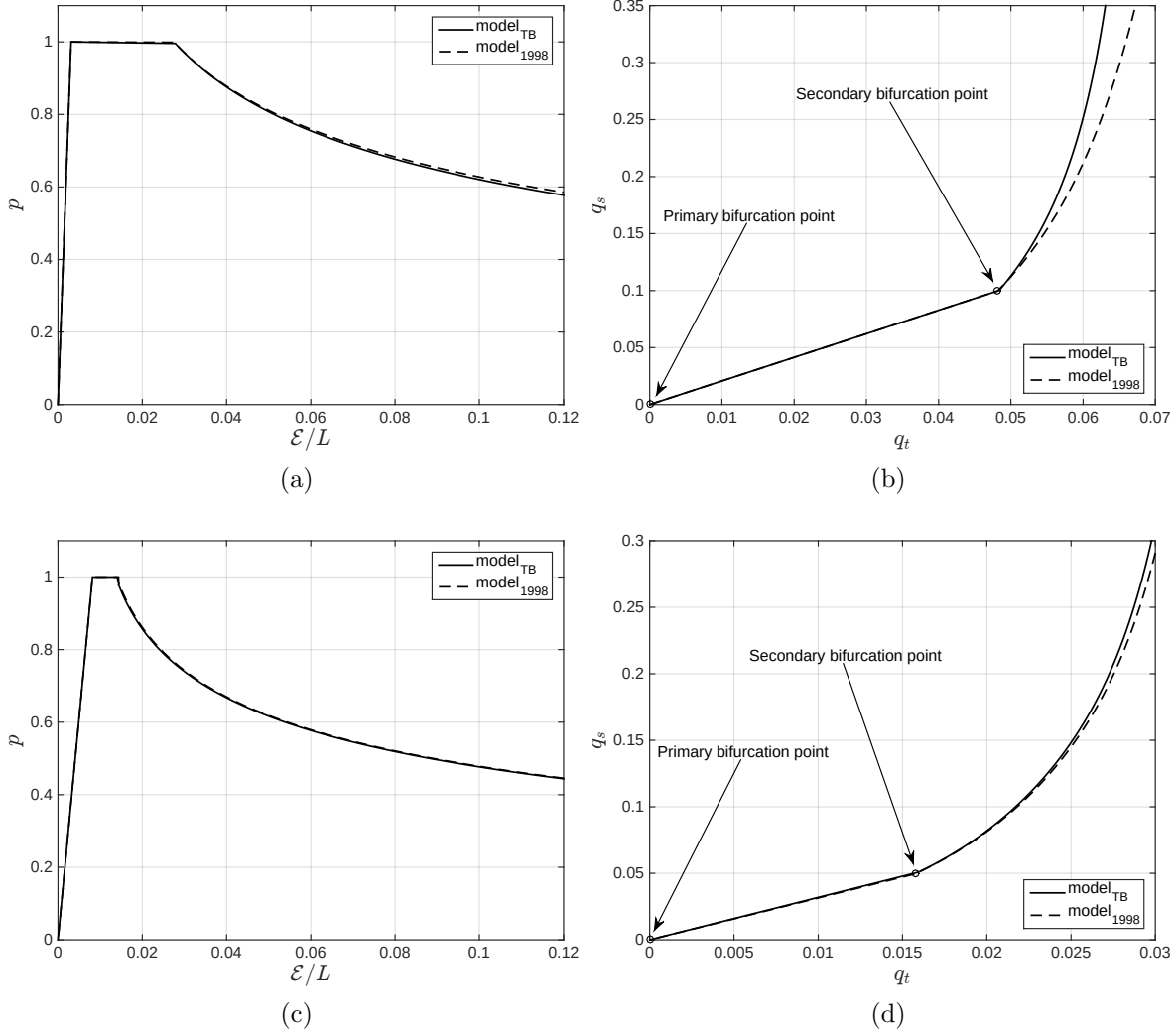


Figure 5: Left: Plotting p versus normalized end shortening (ε/L). Right: Plotting amplitude of tilt q_t versus amplitude of sway q_s for struts of length $L = 100$ mm and different core depths. $b = 5.08$ mm for (a) and (b) and $b = 10.16$ mm for (c) and (d).

the two models, no significant differences can be observed in the post-buckling unloading response beyond the secondary bifurcation. Some very minor differences can be seen in Figure 5(c), which depicts the response of a sandwich strut of core depth $b = 10.16$ mm and length $L = 100$ mm. In both instances, the new model (thereafter termed ‘ model_{TB} ’) has a marginally steeper post-buckling but, nonetheless, following the same unloading trend. A slightly more measureable difference beyond the critical (primary) bifurcation is visible

when comparing the tilt and sway components of the overall mode directly, as shown in Figures 5(b) and (d). These differences, however small, are attributed to the formation of local deformations occurring at the top (less compressed) face plate, which were inhibited within the 1998 model. As seen in Figure 6, which exhibits the evolution of interactive modes w_b , w_t , u_b , u_t with P unloading, as soon as the interactive localized buckle appears within the more compressed face plate, a quasi-periodic buckle pattern appears in the other. The top face plate deformation begins for all configurations as a shallow negative displacement, flattening the top face plate at mid-span and down the post-buckling path, two more half sine waves begin appearing at the two ends. These diagrams show the na-

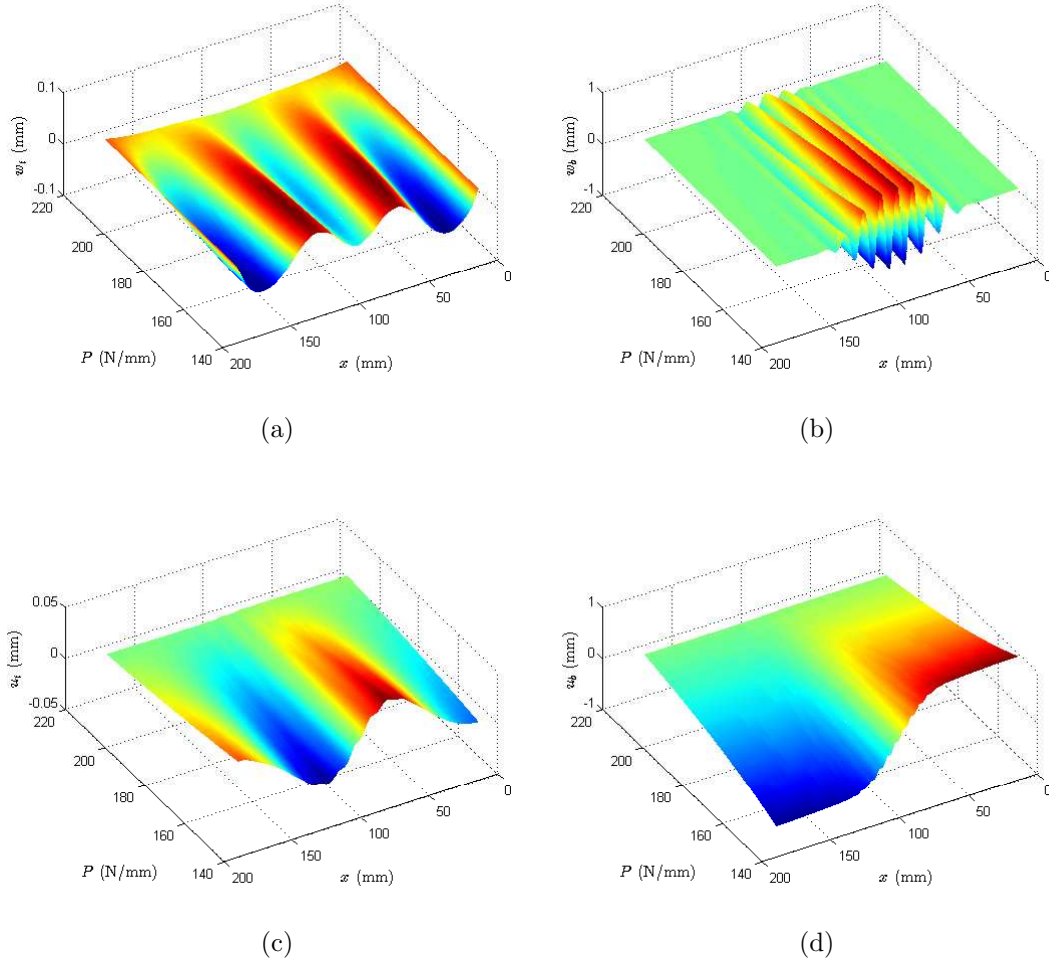


Figure 6: (a) Local mode evolution of face plates unloading for a strut of length $L = 175$ mm and core depth $b = 5.08$ mm. (a) w_t (b) w_b , (c) u_t and (d) u_b .

ture, as well as the comparative magnitude, of the interactive modes. The first observation is that the modes corresponding to the top face plate are smaller in magnitude compared to the ones corresponding to the bottom face plate. This can be attributed to the fact that

the localized interactive buckling displacement occurring within the bottom face plate is a signature of a destabilizing phenomenon and hence grows faster during unloading. The displacement w_t , however small, contributes to the flattening of the top face plate during post-buckling (Figure 7). Parametric studies, while keeping the core depth constant,

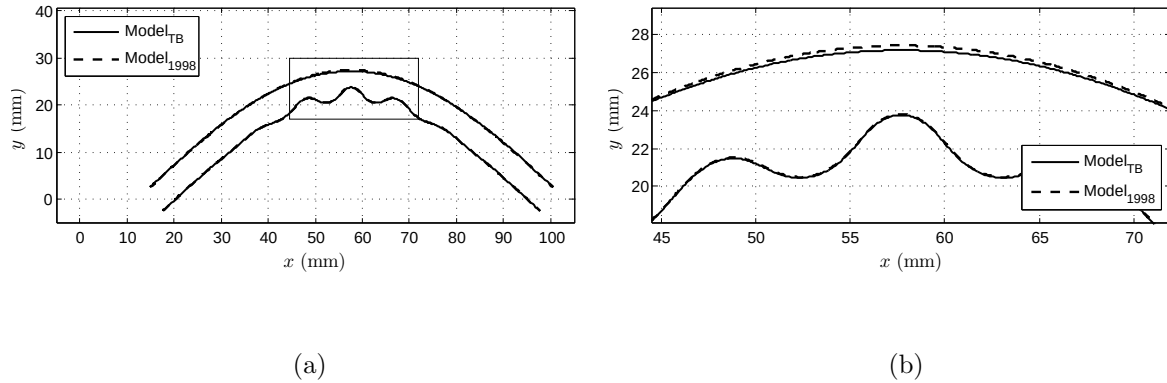


Figure 7: (a) The overall deformations on a $b=5.08$ mm and $L=100$ mm strut at $p=0.7$. (b) A close look at midspan deformations. While the bottom face plate deformations are easily observed, the top face plate local deformations are not as clear but contribute to the flattening of the top face plate.

demonstrate that with increasing length the periodicity in the deformations of the top face plate is maintained, albeit with an increasing number of half sine waves and diminishing growth, as shown in Figure 8. This is expected since with increasing length, more half sine waves would be generated in the most energetically favourable buckle pattern. The amplitude of the less compressed face plate local deformations does not change with increasing length at different p levels. These local deformations are independent of the localized mode that manifests in the more compressed face plate, as shown in Equations (2)-(3). Moreover they have a different wavelength and their amplitude is uniform, unlike the localized mode which is concentrated at midspan. A further parametric study, performed by investigating the amplitude and nature of deformations for different sandwich depths at constant p levels, exhibited an insensitivity to the amplitude of the less compressed face plate deformations; this maintained a constant local displacement amplitude to depth ratio. Conversely, the interactive mode in the more compressed face plate becomes more localized with increasing depth, for the different depths (5.08 to 14 mm). As the load rapidly decreases during post-buckling, local plate buckling at the top does not grow and its effect is quickly diminished due to the tensile stresses from bending introduced by the overall buckling component.

3.1.2. Critical local buckling

An important feature of the model presented currently is that it enables the investigation of interactive localized buckling when local buckling occurs first. The second case considers deeper struts where local buckling becomes critical. Example struts of length

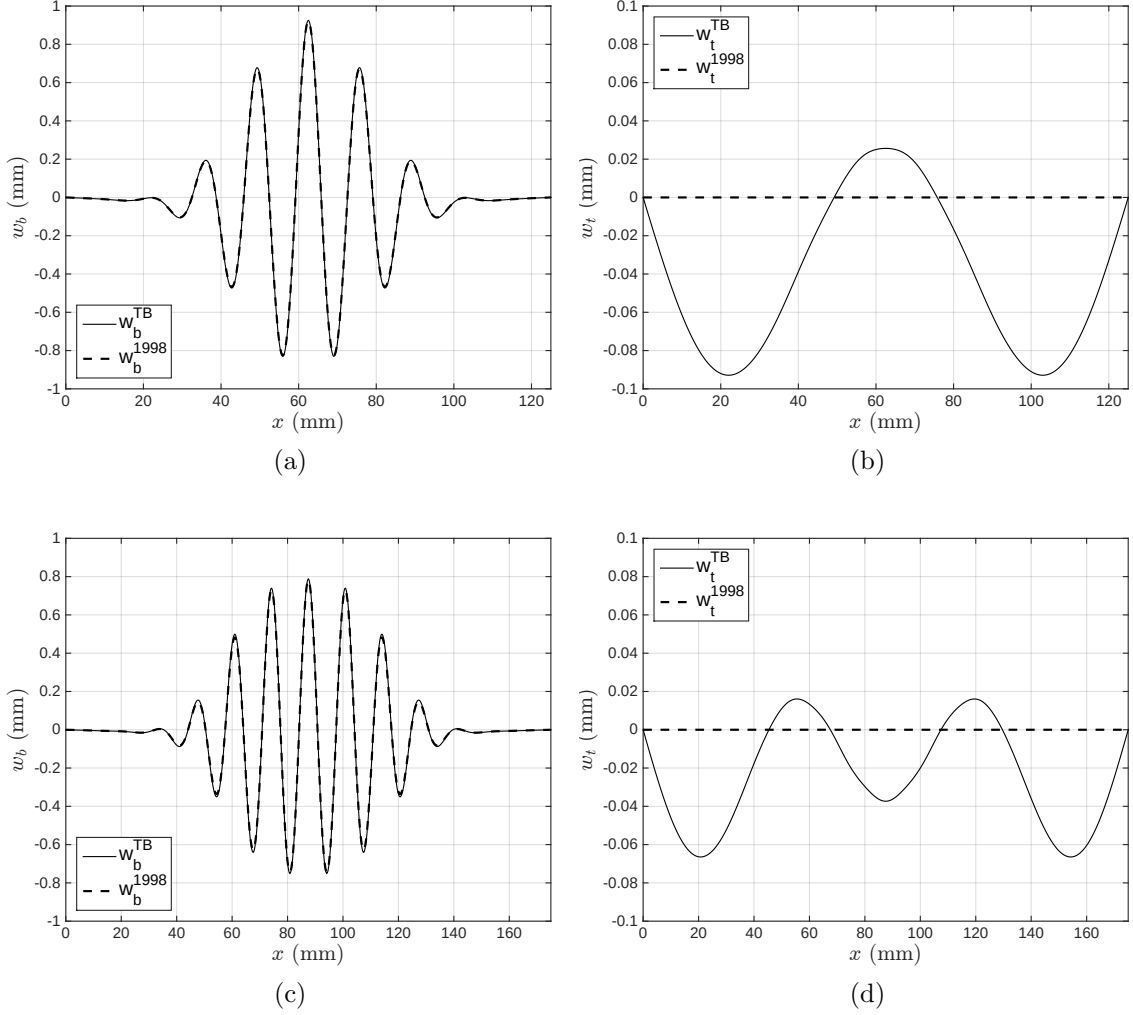


Figure 8: Both face plate local displacements for depth $b = 5.08$ mm. Length $L = 125$ mm for (a–b) and $L = 175$ mm for (c–d). All are shown at $p = 0.70$.

$L = 100$ mm with depths $b = 16$ and 18 mm are analysed. Since the load parameter is normalized with respect to the overall critical load, the primary bifurcations are, in all cases, found to be close to, but below, $p = 1$; specifically p is critical at 0.94 and 0.92 for $b = 16$ mm and $b = 18$ mm respectively. An interesting feature compared to the case where the overall mode is critical, is the continued close proximity of the primary (local mode) and secondary (overall mode) bifurcation. Owing to the close proximity of the two bifurcations, the solver jumped from the primary (local) bifurcation to interactive buckling beyond the secondary bifurcations, thus not revealing the nature (symmetric/antisymmetric) of the local mode that was triggered first. The system subsequently exhibiting a steep destabilizing post-buckling response as the interactive mode is initiated, as observed in Figure 9. The confinement and growth in amplitude of the localized mode in the bottom face plate,

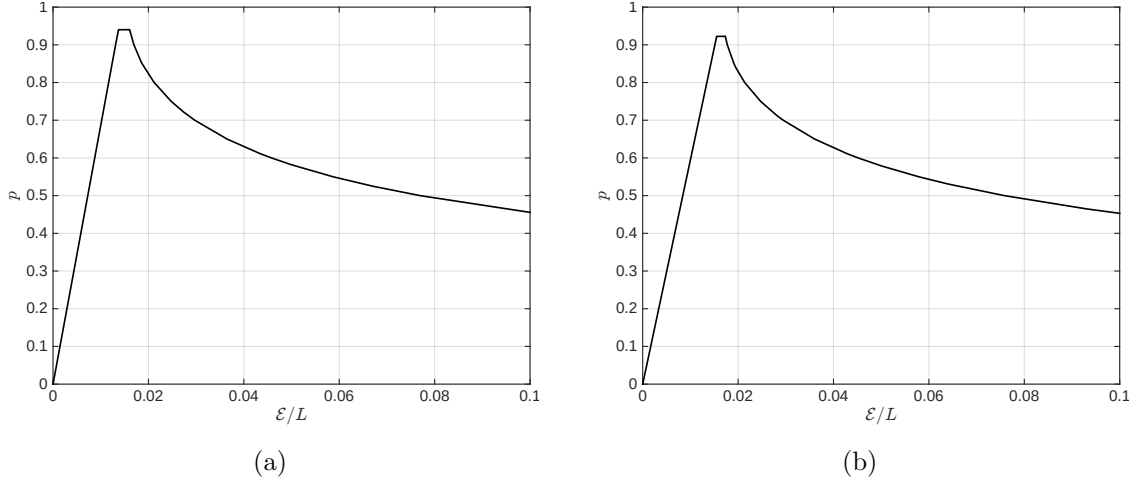


Figure 9: Equilibrium paths for struts with length $L = 100$ mm and core depth (a) $b = 16$ mm, or (b) $b = 18$ mm. The gap $\mathcal{E}/L$ between the primary and secondary bifurcation points is 0.0024 for $b = 16$ mm and 0.0018 for $b = 18$ mm.

as well as the top face plate deformations, lead to a steeper post-buckling response and sudden loss of the load carrying capacity. It has to be noted that these interactive modes cannot be captured by the 1998 model since it does not account for local deformation on both face plates and hence it could not account for the case where local buckling is critical very accurately. However, an attempt was made in [8] to model pure local buckling assuming a mode where the bottom plate deflected with a purely periodic profile; similar findings were made, where the gap between triggering local buckling and then interactive buckling was very small for all geometries that were considered. More differences can be found when examining the mode shapes more closely, the interactive mode observed within the more compressed face plate remains confined to the midspan as the post-buckling response evolves, exhibiting a higher modulation of the main wave, further down the path. This can be attributed to the softer nature of the strut that has a deeper core. On the other hand, local face plate displacements are seen within the top face plate, initially as a sizable negative displacement at midspan of similar amplitude to the localized mode, which then tends to be quasi-periodic, without growing as fast as the localized mode (Figure 10).

4. Comparison with Finite Element simulations

Finite Element (FE) simulations were conducted to capture both the bottom face plate localization as well as the top face plate local deformations. The general finite element software ANSYS was employed to perform the simulations for the buckling and post-buckling response of the sandwich strut in question. Initially, the numerical model was discretized using a two dimensional (2D) solid element PLANE182 which is defined by four nodes and having two degrees of freedom at each node. The mesh was discretized with 1:2 aspect ratio elements and the mesh density of the model was chosen for the instance where the selected and immediately preceding mesh sizes resulted in buckling load solutions that

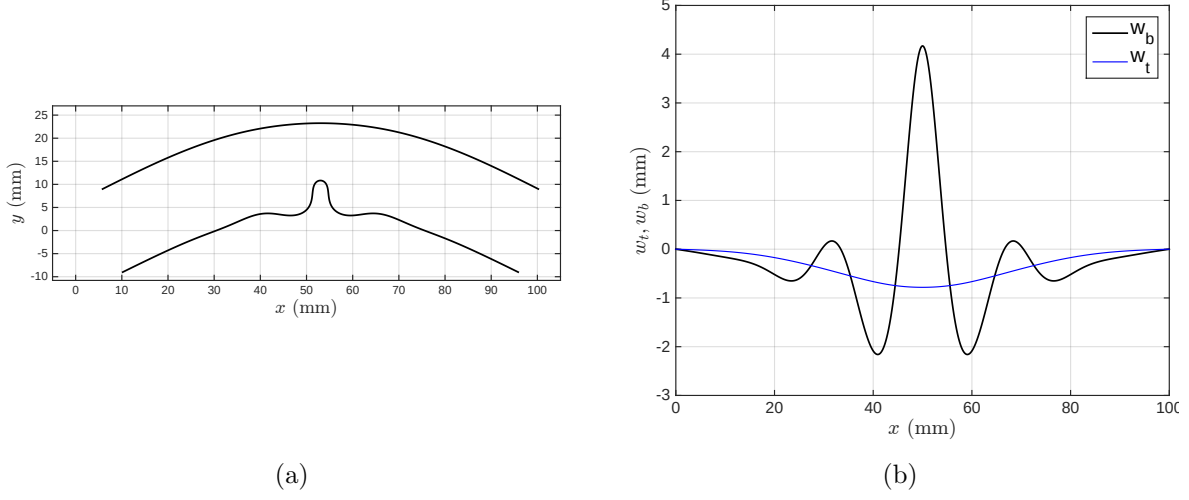


Figure 10: (a) Interactive mode deformation profile for a strut of length $L = 100$ mm and core depth $b = 16$ mm. (b) Local modes within the top and bottom face plate at $\mathcal{E}/L = 10\%$ within the interactive post-buckling range.

were within 0.5% of each other.

In order to match the analytical model as closely as possible, the core depth b' is modelled as $b - t$. The same mesh is used for both the core and the face plates. The left edge elements were free to rotate about a master element (the nodes were prescribed to move along a straight line connecting the two extreme nodes located on the top and bottom faces) at mid-sections which were constrained in the vertical direction (modelling a simple support). On the right edge, simply supported conditions were imposed by restricting horizontal and vertical displacements of the master element, while the remaining elements were free to rotate about the master element. Initially, eigenvalue buckling analysis was performed to calculate buckling loads and mode shapes. As seen in Table 1, the results from FE achieved good agreement when compared to the analytical model in terms of critical loads for both cases of criticality (i.e. when overall buckling is critical or when local buckling is critical). For the cases where local buckling is critical, eigenvalue analysis revealed that

b (mm)	P_{FE}^C (N/mm)	P_{Anal}^C (N/mm)	Mode type
5.08	238.0	218.0	overall (Euler type: $i = 1$)
12.7	807.4	764.8	overall (Euler type: $i = 1$)
16.0	1009.2	959.35	local (Snake: $i = 13$)
18.0	1020.4	1086.8	local (Snake: $i = 13$)

Table 1: Comparison of FE and analytical model results, i represents the number of half-sine waves present in the eigenmode.

the “snake” (antisymmetric) and “hourglass” (symmetric) local modes are extremely close with the “snake” mode having a marginally smaller buckling load, as also observed in the literature [33]. For sufficiently deep cores where local buckling is critical, such as the range

of depths used herein, the strain energy stored at the neutral axis is negligible and remains mostly unaffected, unlike the case of thinner cores where the neutral axis distorts following the buckled pattern of the face plates. Therefore, since the nature of local buckling is mainly governed by the face plate stiffness and thickness, as well as the core stiffness, the critical loads for the antisymmetric (snake) and symmetric (hourglass) modes of the same number of half-sine waves tend to be very close.

The analytical model solutions generated in AUTO did not reveal whether the anti-symmetric or symmetric path was followed, since the close proximity of the primary and secondary bifurcations led to a jump from the primary bifurcation on the fundamental path to the secondary bifurcation at the onset of interactive buckling. A Rayleigh-Ritz model, developed in [3], which is based on potential energy principles, was utilized to find the critical loads of the most energetically favourable periodic modes. The critical load and number of half-sine waves for the symmetric (hourglass) modes were found through eigenvalue analysis. Both the number of half-sine waves and the critical loads matched the ones found from FE simulations and the critical modes in AUTO, hence providing confidence in the analytical results. Using this Rayleigh–Ritz model, the antisymmetric mode yielded a much higher critical load due to the assumed buckling mode displacements that considered distortion of the neutral axis, which would be expected for sandwich panels with a low b/t ratio. This assumption was valid within the context of [3], but it would not apply for the higher b/t ratios used herein and required for local buckling to be critical.

For post-buckling analysis, nonlinear geometry effects are turned on and the response of the strut is evaluated beyond the critical bifurcation. The nonlinear analysis is performed using displacement control and utilizes the arclength method in order to capture the nonlinear response beyond the primary and secondary bifurcations. Geometric imperfections are applied for geometrically nonlinear analysis, in order to avoid the bifurcation points and trace an imperfect path around them. In this case, imperfections are assumed to be affine to the lowest (critical) buckling mode obtained from eigenvalue analysis for each case. The amplitude of the applied imperfection is selected to be $L/10000$ which has been demonstrated to be sufficient to trigger the instability and trace the post-buckling equilibrium path as closely as possible to the perfect case, without significant deviations from the perfect case.

The equilibrium diagrams compare very well with the analytical model results for different strut depths, as seen in Figure 11. There was good agreement for the critical loads, which improved for deeper struts. The difference in the overall buckling loads results from the simplified kinematics in the analytical model for the overall mode, since the primary objective is to capture interactive buckling and is more appropriate for longer struts where the buckling loads for symmetric and antisymmetric modes are sufficiently close together. The agreement is even better in the post-buckling range, where the responses follow the same (unstable) unloading trend. For the case where overall buckling is triggered first, the principal difference can be seen in the far post-buckling range. The FE simulation locks the localized mode within the bottom face plate at midspan, which grows as the load reduces during unstable post-buckling, while for the analytical model, progressing down the post-buckling path, the localized response tends to resemble cellular behaviour somewhat

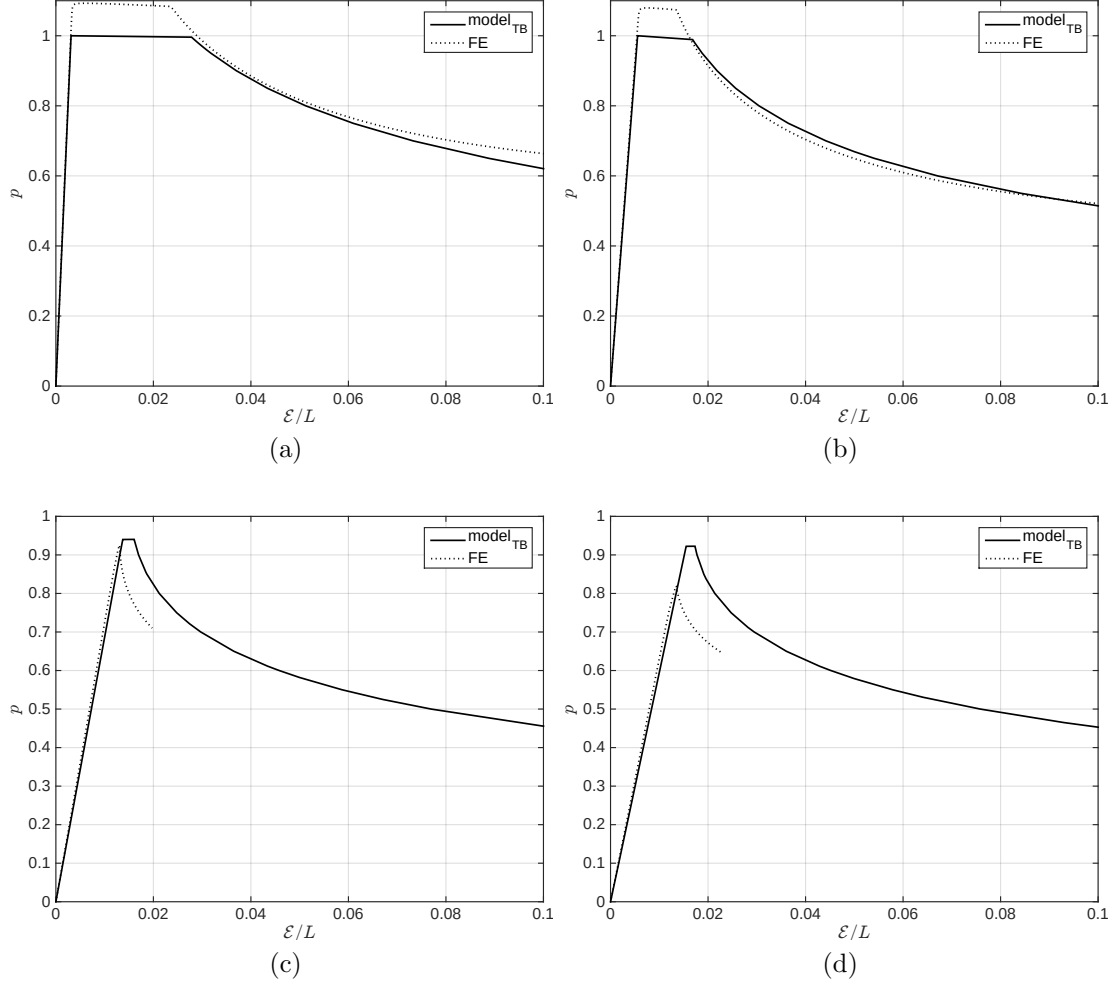


Figure 11: Comparison of equilibrium paths for different struts of length $L = 100$ mm and core depths of: (a) $b = 5.08$ mm, (b) $b = 7.62$ mm, (c) $b = 16$ mm and (d) $b = 18$ mm. (a–b) Overall buckling is critical; (c–d) Local buckling is critical.

with localized waves spreading towards the edges [28, 34, 35, 36].

For the case where local buckling is critical the FE model generated equilibrium paths that completely bypassed the primary bifurcation and headed directly to the secondary bifurcation which leads to unloading, which can be attributed to the use of geometrical imperfections in the nonlinear analysis. Even though this also applies to the critical overall buckling case, the extremely close proximity of the two bifurcation points in the case where local buckling is critical makes the effect more pronounced. The close proximity of the primary and secondary bifurcations when local buckling is critical, has also been observed in the work of Le Grogne and Saoud [37] who studied nonlinear post-buckling of sandwich struts, for differing boundary conditions. Nevertheless, the unloading pattern of the post-buckling path are in good agreement in terms of the rate of unloading, with the two curves

being essentially parallel.

The deformed shapes at equal normalized end shortening also agree very well, the wavelength and the amplitude of localization match well for the bottom face plate, as shown in Figure 12(a). The numerical simulations in all cases exhibited local deformations on the top face plate, diverging from the classical overall eigenmode with associated localized buckling in the bottom face plate. To evaluate the amplitude and nature of the local deformations on either face plate approximately, a half-sine wave is subtracted from the top and bottom face plate deformations. For the top face plate, the local deformations are mostly quasi-periodic in nature, starting with a negative displacement and later forming one or more crests, depending on the length of the strut. The main difference between the numerical simulations and the analytical model is that, in the far post-buckling range, the top face plate deformations in numerical simulations seem to be influenced by the bottom face plate localized mode, with a kink appearing at midspan, as shown in Figure 12(b–c). This phenomenon is deliberately not modelled in the analytical model such that the model remains fairly simple in concept, otherwise non-local effects may become important, with one face plate strongly influencing the other, while the large displacement analysis in the finite element simulation, locks the localized mode in the more compressed face plate to such an extent that it essentially affects the less compressed face plate, at midspan. In the absence of this, away from midspan, the deformations seem to agree well both in nature and magnitude. For deeper struts, where local buckling is critical, there is also excellent agreement between the analytical model and the FE simulations in terms of the confinement of the localized mode at midspan, as seen in Figure 13. The difference in the amplitude of sway, observed in Figure 13(a), can be attributed to the difference in end shortening between the equilibrium paths at the same load level. Looking in detail at the decomposed local deformations, the numerical simulations are in very good agreement with the analytical results in terms of wavelengths. The only differences observed are some smaller amplitude higher frequency waves in the FE simulations, as seen in Figure 13(c), which may be attributed to the local imperfection that was used to initiate the nonlinear analysis, as well as the propagation of the more compressed face plate interactive mode, which contaminates the response. In terms of the amplitudes, some of the less stiff behaviour of the overall panel and the bottom face plate can be attributed to the neglecting of the axial stiffness of the core, which increases in significance for panels with deeper cores.

5. Conclusions

A variational formulation based on potential energy principles accounting for interactive buckling in sandwich struts has been presented. The model builds on the fundamental work by Hunt & Wadee [4] and captures local quasi-periodic deformations within the top, less compressed, face plate in the post-buckling range. The deformations in the less compressed face plate have a different wavelength to the interactive buckle in the more compressed face plate; both grow with progressive unloading in the post-buckling range, but the deformations in the less compressed face plate are always significantly smaller in amplitude than the localized interactive mode that dominates the response. Indeed, the

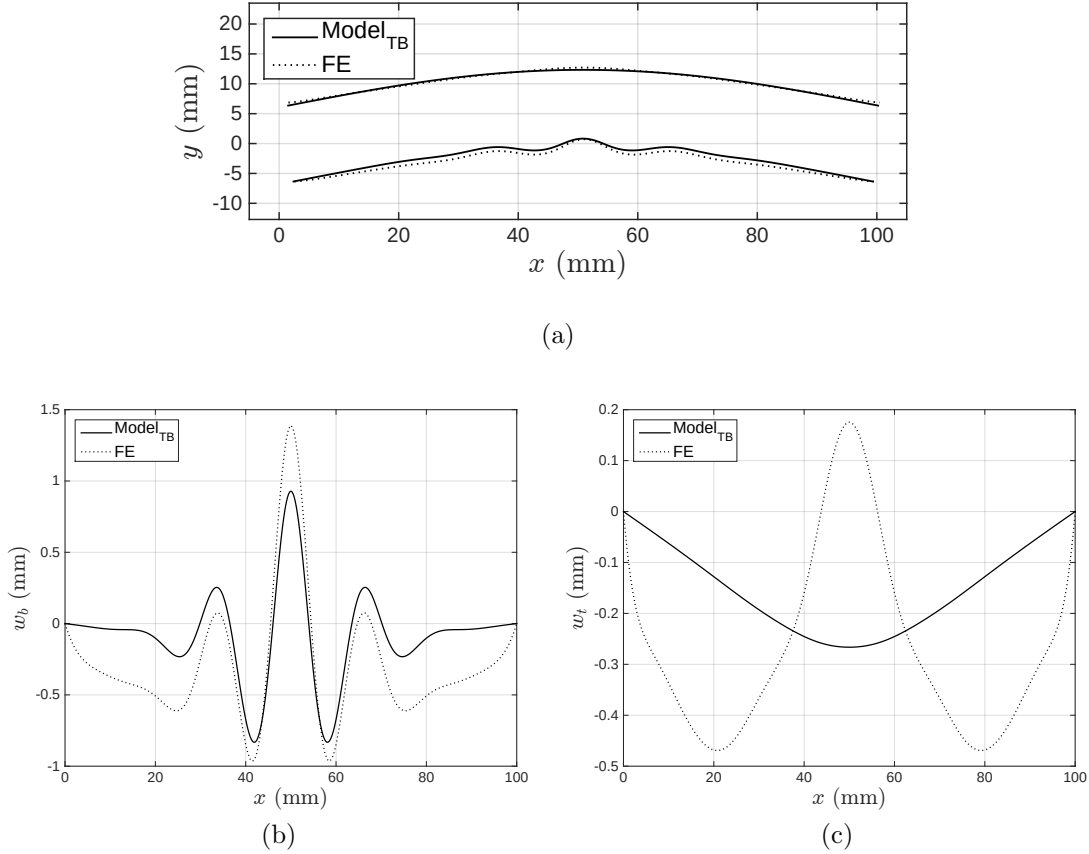


Figure 12: Comparison of deformation profiles from the analytical model and FE simulations for a strut of depth $b = 12.7$ mm and length $L = 100$ mm at $\varepsilon/L = 2.2\%$. (a) Total deformations; (b) bottom face plate local deformations w_b and (c) top face plate local deformations w_t .

dominance of the interactive buckling mode is highlighted by the minimal difference in the equilibrium paths, showing that the local deformations in the less compressed face plate play a negligible influence in the mechanical response of the sandwich strut, for the thickness considered and where overall buckling was critical. More importantly for the current discussion, however, the model is capable of dealing with the case where local buckling is critical. This case exhibits a very close proximity between the primary and secondary bifurcations and a severely unstable response is the result, which would translate into a high degree of imperfection sensitivity [8]. These results show good agreement with the FE results which also capture the inhomogeneous quasi-periodic deformations in the top (less compressed) face plate.

This formulation is seen as an intermediate step in the study of interactive buckling in sandwich struts. Further work will seek to extend model to include other boundary conditions [37], imperfections, other loading combinations and more importantly slip interfaces which will enable the capture of delaminations [6, 7] and their propagation on either face

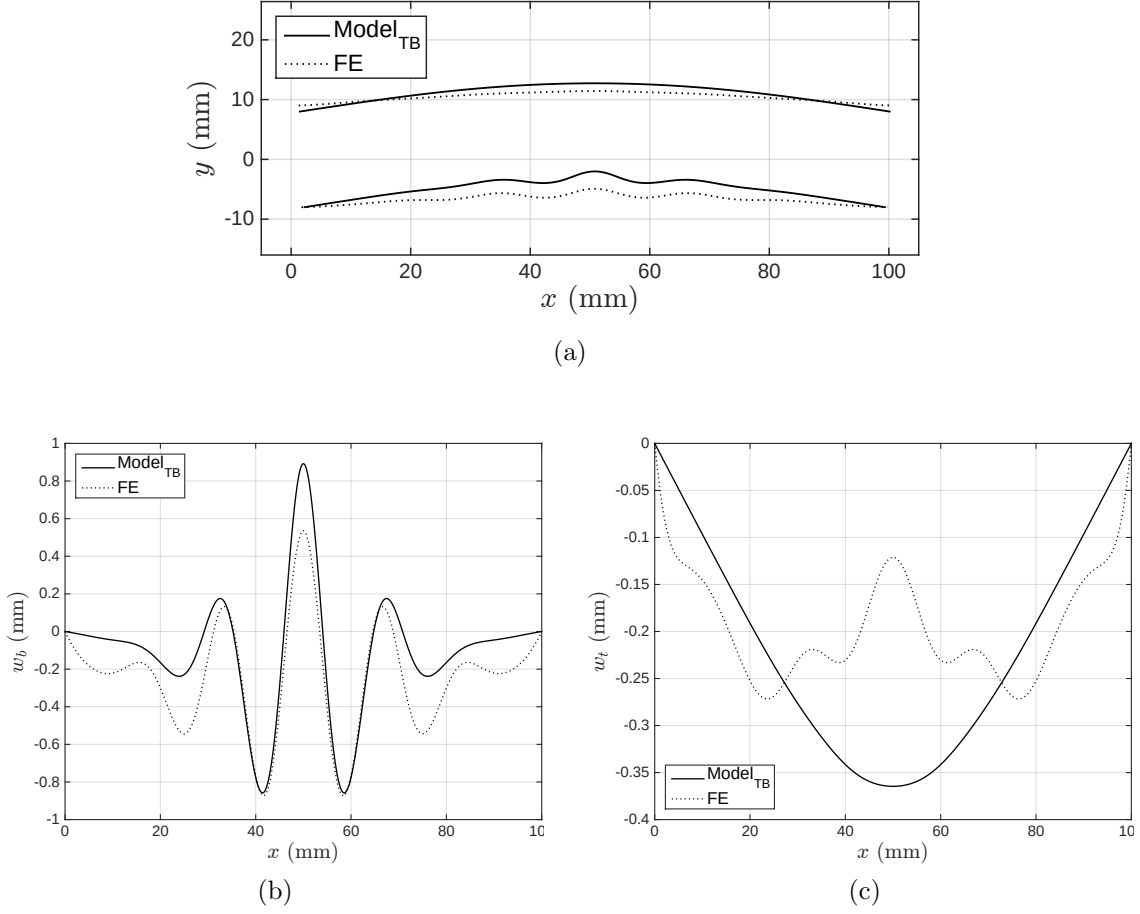


Figure 13: Comparison of deformation profiles from the analytical model and FE simulations for a strut of depth $b = 16$ mm and length $L = 100$ mm at $p = 0.80$. (a) Total deformations; (b) bottom face plate local deformations w_b and (c) top face plate local deformations w_t . Note that the FE result in (a) is possibly affected by the chosen imperfection used to trigger buckling in the nonlinear analysis.

plate. Finally the authors are planning the development of experimental tests to capture interactive buckling and inhomogeneous deformations in the panels using Digital Image Correlation (DIC) techniques through photogrammetric methods [38].

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